CSE370 : Database Systems

Assignment 02 | Spring 26

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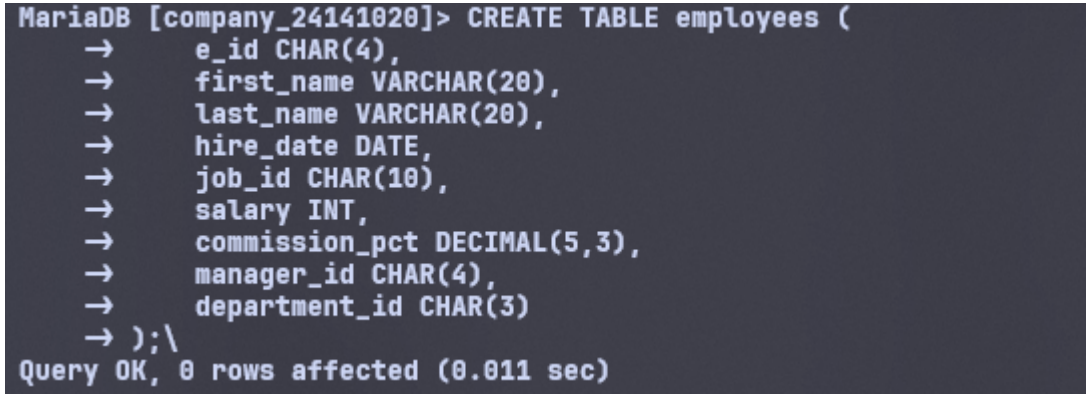
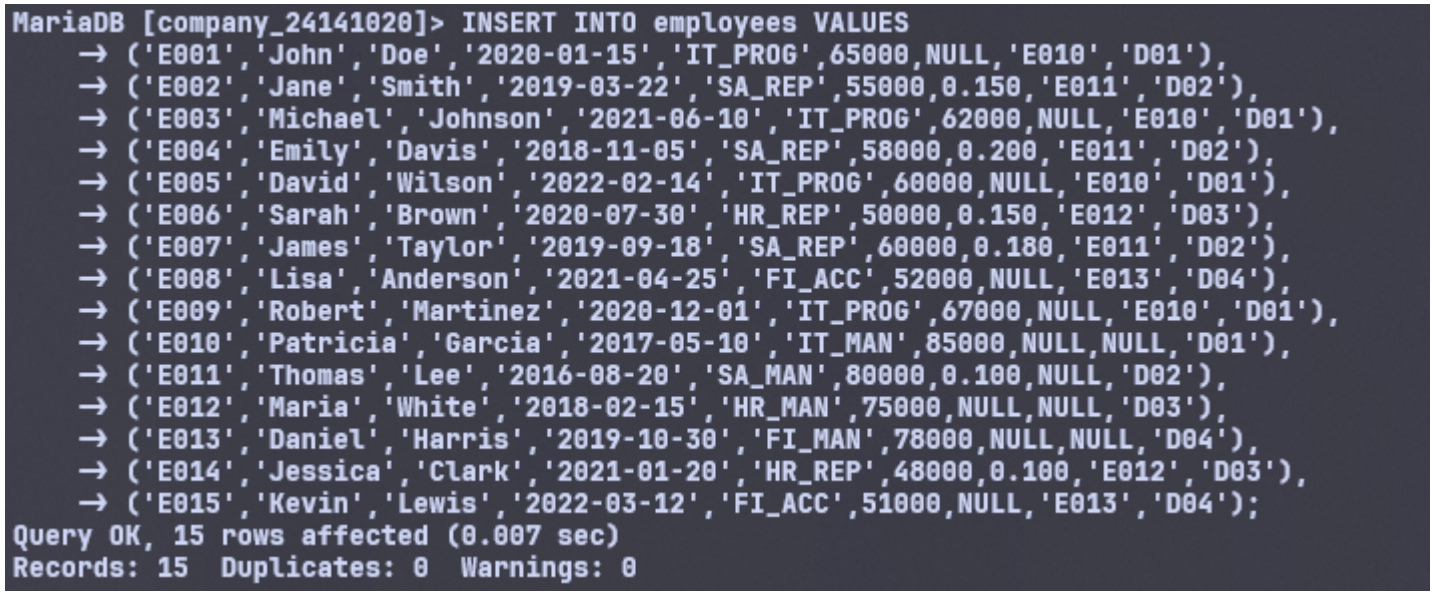
No 1 Query (as Plain Text)	CREATE TABLE employees (e_id CHAR(4), first_name VARCHAR(20), last_name VARCHAR(20), hire_date DATE, job_id CHAR(10), salary INT, commission_pct DECIMAL(5,3), manager_id CHAR(4), department_id CHAR(3));
No 1 SS (of Query & Output in Shell)	 MariaDB [company_24141020]> CREATE TABLE employees (-> e_id CHAR(4), -> first_name VARCHAR(20), -> last_name VARCHAR(20), -> hire_date DATE, -> job_id CHAR(10), -> salary INT, -> commission_pct DECIMAL(5,3), -> manager_id CHAR(4), -> department_id CHAR(3) ->);\ Query OK, 0 rows affected (0.011 sec)

```
MariaDB [company_24141020]> describe employees;
+----------------+--------------+------+-----+---------+-------+
| Field          | Type         | Null | Key | Default | Extra |
+----------------+--------------+------+-----+---------+-------+
| e_id           | char(4)      | YES  |     | NULL    |       |
| first_name     | varchar(20)  | YES  |     | NULL    |       |
| last_name      | varchar(20)  | YES  |     | NULL    |       |
| hire_date      | date         | YES  |     | NULL    |       |
| job_id         | char(10)     | YES  |     | NULL    |       |
| salary         | int(11)      | YES  |     | NULL    |       |
| commission_pct | decimal(5,3) | YES  |     | NULL    |       |
| manager_id     | char(4)      | YES  |     | NULL    |       |
| department_id  | char(3)      | YES  |     | NULL    |       |
+----------------+--------------+------+-----+---------+-------+
9 rows in set (0.007 sec)
```

No 2 Query
(as Plain Text)

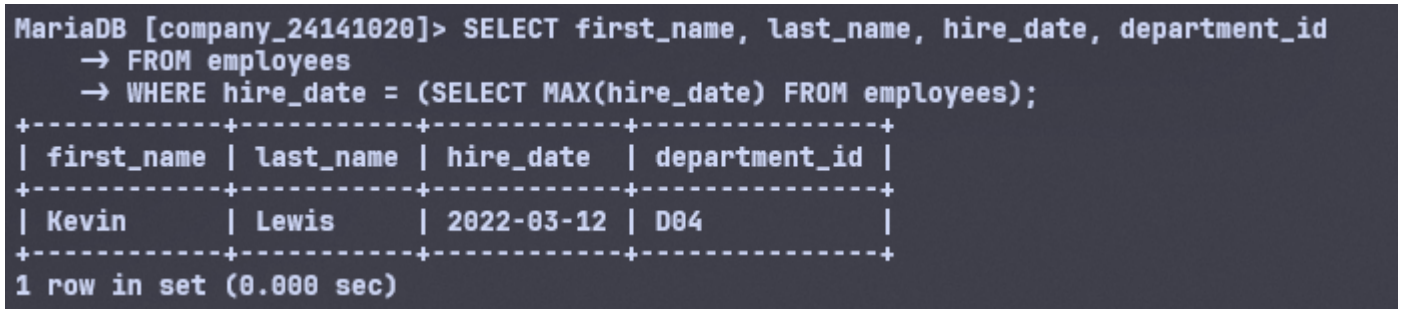
INSERT INTO employees VALUES
('E001','John','Doe','2020-01-15','IT_PROG',65000,NULL,'E010','D01'),
('E002','Jane','Smith','2019-03-22','SA_REP',55000,0.150,'E011','D02'),
('E003','Michael','Johnson','2021-06-10','IT_PROG',62000,NULL,'E010','D01'),
('E004','Emily','Davis','2018-11-05','SA_REP',58000,0.200,'E011','D02'),
('E005','David','Wilson','2022-02-14','IT_PROG',60000,NULL,'E010','D01'),
('E006','Sarah','Brown','2020-07-30','HR_REP',50000,0.150,'E012','D03'),
('E007','James','Taylor','2019-09-18','SA_REP',60000,0.180,'E011','D02'),
('E008','Lisa','Anderson','2021-04-25','FI_ACC',52000,NULL,'E013','D04'),
('E009','Robert','Martinez','2020-12-01','IT_PROG',67000,NULL,'E010','D01'),
('E010','Patricia','Garcia','2017-05-10','IT_MAN',85000,NULL,NULL,'D01'),
('E011','Thomas','Lee','2016-08-20','SA_MAN',80000,0.100,NULL,'D02'),
('E012','Maria','White','2018-02-15','HR_MAN',75000,NULL,NULL,'D03'),
('E013','Daniel','Harris','2019-10-30','FI_MAN',78000,NULL,NULL,'D04'),

	('E014','Jessica','Clark','2021-01-20','HR_REP',48000,0.100,'E012','D03'), ('E015','Kevin','Lewis','2022-03-12','FI_ACC',51000,NULL,'E013','D04');
No 2 SS **(of Query & Output in Shell)**	

```
MariaDB [company_24141020]> INSERT INTO employees VALUES
    -> ('E001','John','Doe','2020-01-15','IT_PROG',65000,NULL,'E010','D01'),
    -> ('E002','Jane','Smith','2019-03-22','SA_REP',55000,0.150,'E011','D02'),
    -> ('E003','Michael','Johnson','2021-06-10','IT_PROG',62000,NULL,'E010','D01'),
    -> ('E004','Emily','Davis','2018-11-05','SA_REP',58000,0.200,'E011','D02'),
    -> ('E005','David','Wilson','2022-02-14','IT_PROG',60000,NULL,'E010','D01'),
    -> ('E006','Sarah','Brown','2020-07-30','HR_REP',50000,0.150,'E012','D03'),
    -> ('E007','James','Taylor','2019-09-18','SA_REP',60000,0.180,'E011','D02'),
    -> ('E008','Lisa','Anderson','2021-04-25','FI_ACC',52000,NULL,'E013','D04'),
    -> ('E009','Robert','Martinez','2020-12-01','IT_PROG',67000,NULL,'E010','D01'),
    -> ('E010','Patricia','Garcia','2017-05-10','IT_MAN',85000,NULL,NULL,'D01'),
    -> ('E011','Thomas','Lee','2016-08-20','SA_MAN',80000,0.100,NULL,'D02'),
    -> ('E012','Maria','White','2018-02-15','HR_MAN',75000,NULL,NULL,'D03'),
    -> ('E013','Daniel','Harris','2019-10-30','FI_MAN',78000,NULL,NULL,'D04'),
    -> ('E014','Jessica','Clark','2021-01-20','HR_REP',48000,0.100,'E012','D03'),
    -> ('E015','Kevin','Lewis','2022-03-12','FI_ACC',51000,NULL,'E013','D04');
Query OK, 15 rows affected (0.007 sec)
Records: 15  Duplicates: 0  Warnings: 0
```

```
MariaDB [company_24141020]> SELECT * FROM employees;
+------+------------+-----------+------------+---------+--------+----------------+------------+---------------+
| e_id | first_name | last_name | hire_date  | job_id  | salary | commission_pct | manager_id | department_id |
+------+------------+-----------+------------+---------+--------+----------------+------------+---------------+
| E001 | John       | Doe       | 2020-01-15 | IT_PROG |  65000 |           NULL | E010       | D01           |
| E002 | Jane       | Smith     | 2019-03-22 | SA_REP  |  55000 |          0.150 | E011       | D02           |
| E003 | Michael    | Johnson   | 2021-06-10 | IT_PROG |  62000 |           NULL | E010       | D01           |
| E004 | Emily      | Davis     | 2018-11-05 | SA_REP  |  58000 |          0.200 | E011       | D02           |
| E005 | David      | Wilson    | 2022-02-14 | IT_PROG |  60000 |           NULL | E010       | D01           |
| E006 | Sarah      | Brown     | 2020-07-30 | HR_REP  |  50000 |          0.150 | E012       | D03           |
| E007 | James      | Taylor    | 2019-09-18 | SA_REP  |  60000 |          0.180 | E011       | D02           |
| E008 | Lisa       | Anderson  | 2021-04-25 | FI_ACC  |  52000 |           NULL | E013       | D04           |
| E009 | Robert     | Martinez  | 2020-12-01 | IT_PROG |  67000 |           NULL | E010       | D01           |
| E010 | Patricia   | Garcia    | 2017-05-10 | IT_MAN  |  85000 |           NULL | NULL       | D01           |
| E011 | Thomas     | Lee       | 2016-08-20 | SA_MAN  |  80000 |          0.100 | NULL       | D02           |
| E012 | Maria      | White     | 2018-02-15 | HR_MAN  |  75000 |           NULL | NULL       | D03           |
| E013 | Daniel     | Harris    | 2019-10-30 | FI_MAN  |  78000 |           NULL | NULL       | D04           |
| E014 | Jessica    | Clark     | 2021-01-20 | HR_REP  |  48000 |          0.100 | E012       | D03           |
| E015 | Kevin      | Lewis     | 2022-03-12 | FI_ACC  |  51000 |           NULL | E013       | D04           |
+------+------------+-----------+------------+---------+--------+----------------+------------+---------------+
15 rows in set (0.000 sec)
```

No 3 Query (as Plain Text)	SELECT first_name, last_name, hire_date, department_id FROM employees WHERE hire_date = (SELECT MAX(hire_date) FROM employees);

No 3 SS (of Query & Output in Shell)	
No 4 Query (as Plain Text)	SELECT first_name, last_name, e_id, salary, department_id FROM employees e1 WHERE salary = (SELECT MAX(salary) FROM employees e2 WHERE e2.department_id = e1.department_id);

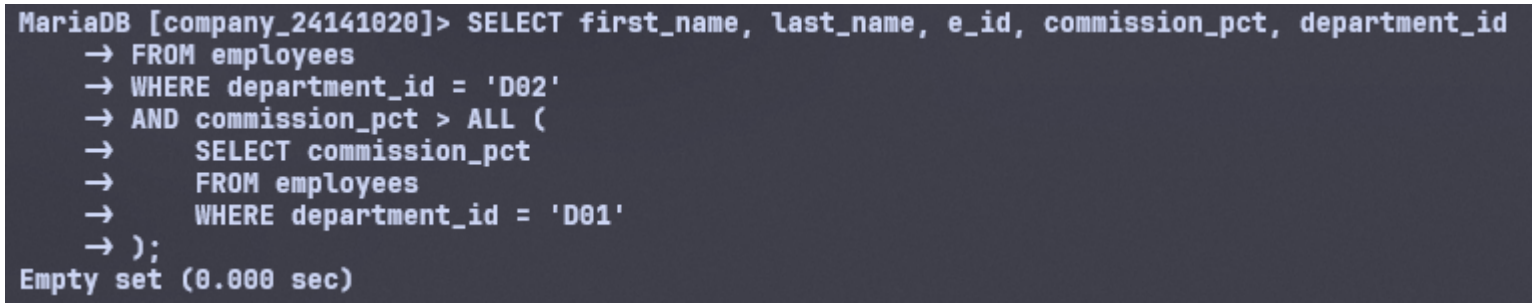
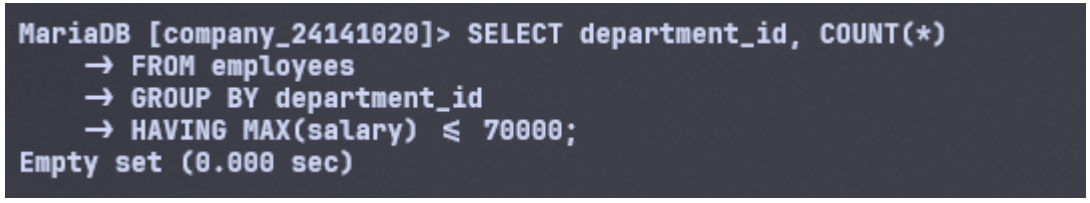
No 4 SS (of Query & Output in Shell)	

```
unknown [company_24141020]> SELECT first_name, last_name, e_id, salary, department_id
    -> FROM employees e1
    -> WHERE salary = (
    ->     SELECT MAX(salary)
    ->     FROM employees e2
    ->     WHERE e2.department_id = e1.department_id
    -> );
No connection. Trying to reconnect...
Connection id:    8
Current database: company_24141020

+------------+-----------+------+--------+---------------+
| first_name | last_name | e_id | salary | department_id |
+------------+-----------+------+--------+---------------+
| Patricia   | Garcia    | E010 |  85000 | D01           |
| Thomas     | Lee       | E011 |  80000 | D02           |
| Maria      | White     | E012 |  75000 | D03           |
| Daniel     | Harris    | E013 |  78000 | D04           |
+------------+-----------+------+--------+---------------+
4 rows in set (0.002 sec)
```

No 5 Query (as Plain Text)	

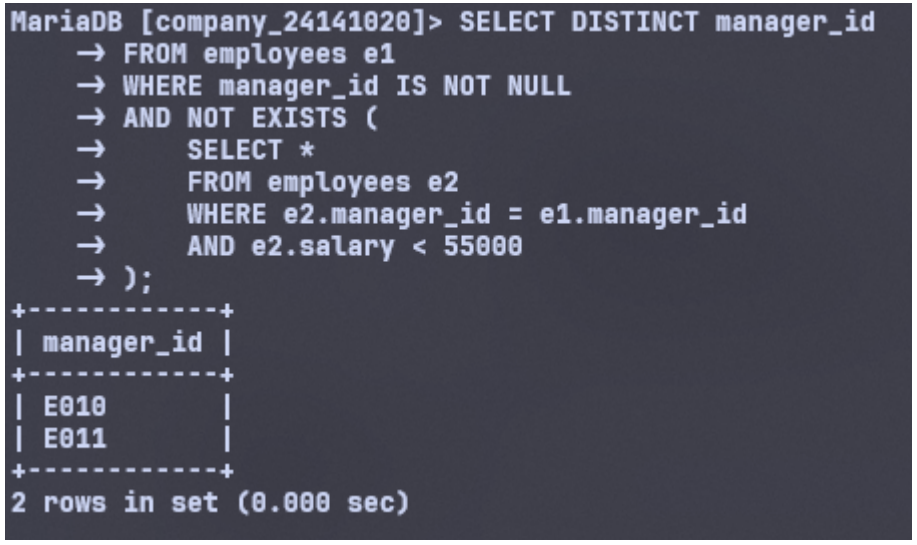
```
SELECT first_name, last_name, e_id, commission_pct, department_id
FROM employees
WHERE department_id = 'D02'
AND commission_pct > ALL (
    SELECT commission_pct
    FROM employees
    WHERE department_id = 'D01'
);
```

No 5 SS (of Query & Output in Shell)	
No 6 Query (as Plain Text)	SELECT department_id, COUNT(*) FROM employees GROUP BY department_id HAVING MAX(salary) <= 70000;
No 6 SS (of Query & Output in Shell)	
No 7 Query (as Plain Text)	SELECT department_id, job_id, salary FROM employees e1 WHERE salary < ANY (SELECT salary FROM employees e2 WHERE e2.department_id = e1.department_id);

No 7 SS (of Query & Output in Shell)	

```
MariaDB [company_24141020]> SELECT department_id, job_id, salary
    -> FROM employees e1
    -> WHERE salary < ANY (
    ->     SELECT salary
    ->     FROM employees e2
    ->     WHERE e2.department_id = e1.department_id
    -> );
+---------------+---------+--------+
| department_id | job_id  | salary |
+---------------+---------+--------+
| D01           | IT_PROG |  65000 |
| D02           | SA_REP  |  55000 |
| D01           | IT_PROG |  62000 |
| D02           | SA_REP  |  58000 |
| D01           | IT_PROG |  60000 |
| D03           | HR_REP  |  50000 |
| D02           | SA_REP  |  60000 |
| D04           | FI_ACC  |  52000 |
| D01           | IT_PROG |  67000 |
| D03           | HR_REP  |  48000 |
| D04           | FI_ACC  |  51000 |
+---------------+---------+--------+
11 rows in set (0.000 sec)
```

No 8 Query (as Plain Text)	SELECT DISTINCT manager_id FROM employees e1 WHERE manager_id IS NOT NULL

	AND NOT EXISTS (SELECT * FROM employees e2 WHERE e2.manager_id = e1.manager_id AND e2.salary < 55000);
No 8 SS (of Query & Output in Shell)	 MariaDB [company_24141020]> SELECT DISTINCT manager_id → FROM employees e1 → WHERE manager_id IS NOT NULL → AND NOT EXISTS (→ SELECT * → FROM employees e2 → WHERE e2.manager_id = e1.manager_id → AND e2.salary < 55000 →); +----------+ | manager_id | +----------+ | E010 | | E011 | +----------+ 2 rows in set (0.000 sec)
No 9 Query (as Plain Text)	SELECT first_name, last_name, e_id, salary, department_id, commission_pct FROM employees e1 WHERE commission_pct = (

	SELECT MAX(commission_pct) FROM employees e2 WHERE e2.manager_id = e1.manager_id);
No 9 SS (of Query & Output in Shell)	

```
MariaDB [company_24141020]> SELECT first_name, last_name, e_id, salary, department_id, commission_pct
    -> FROM employees e1
    -> WHERE commission_pct = (
    ->     SELECT MAX(commission_pct)
    ->     FROM employees e2
    ->     WHERE e2.manager_id = e1.manager_id
    -> );
+------------+-----------+------+--------+---------------+----------------+
| first_name | last_name | e_id | salary | department_id | commission_pct |
+------------+-----------+------+--------+---------------+----------------+
| Emily      | Davis     | E004 |  58000 | D02           |          0.200 |
| Sarah      | Brown     | E006 |  50000 | D03           |          0.150 |
+------------+-----------+------+--------+---------------+----------------+
2 rows in set (0.000 sec)
```

No 10 Query (as Plain Text)	SELECT job_id, AVG(salary) FROM employees GROUP BY job_id HAVING AVG(salary) < (SELECT AVG(salary) FROM employees);

No 10 SS (of Query & Output in Shell)	

```
MariaDB [company_24141020]> SELECT job_id, AVG(salary)
    -> FROM employees
    -> GROUP BY job_id
    -> HAVING AVG(salary) < (SELECT AVG(salary) FROM employees);
+--------+-------------+
| job_id | AVG(salary) |
+--------+-------------+
| FI_ACC |  51500.0000 |
| HR_REP |  49000.0000 |
| SA_REP |  57666.6667 |
+--------+-------------+
3 rows in set (0.000 sec)
```